

From three glass plates to one photograph

CS180 Project 1 - robust reconstruction with phase correlation, local NCC refinement, overlap-aware cropping, and color cleanup

The final pipeline separates the vertical B / G / R negatives, estimates integer translations without circular wrap-around, and reports both NCC and normalized L2 error. The two images below are representative results included for visual inspection.



Emir



self_portrait

Submission contents

The accompanying HTML report contains all 14 course samples, a separate LOC section with three self-selected collection examples, the full offset/crop records, an optimization history, and direct provenance links. The displayed galleries are ordered by the recorded quality summary from higher NCC to lower NCC.

Algorithm progression: V0 single-scale NCC -> V1 pyramid search -> V2 FFT phase initialization plus local NCC/edge refinement -> V3 consecutive-run border cleanup and normalized L2 audit.

Method and optimization log

Every input is treated with the same parameters. The blue plate is the reference; green and red are translated by (dy, dx). Candidate scores are evaluated only on true overlap pixels.

$$\text{score} = 0.70 * \text{NCC}(\text{intensity}) + 0.30 * \text{NCC}(\text{edge magnitude})$$

Large TIFFs first use a Hann-windowed FFT cross-power spectrum on a 4x edge thumbnail. A small full-resolution search then refines the peak. The output is intersected across all translated channels; a data-dependent border detector removes clipped black/white and cross-channel registration strips. A 1st/99th percentile contrast stretch, restrained gray-world gains, and mild gamma produce the displayed RGB image.

Metrics

NCC is mean-subtracted normalized cross-correlation (higher is better). L2 is the RMSE after z-normalizing the two overlapping plates (lower is better), so exposure differences do not dominate the alignment audit. Both values are recorded separately for G -> B and R -> B.

version	change	observed effect
V0	Exhaustive single-scale NCC on $[-15, 15]^2$	Simple baseline; too small for large TIFF shifts
V1	Custom 2x pyramid plus intensity/edge NCC	Fast, but coarse alias peaks remained on Emir
V2	FFT phase initialization plus local refinement	Removed large yellow/purple channel ghosts
V3	Overlap crop, consecutive good-run test, L2 audit	Removed scan margins and made quality measurable

Required JPEG single-scale baseline

The three low-resolution JPEGs are also evaluated with the required exhaustive NCC/L2 search. The pyramid output is the final result; this table preserves the baseline measurements.

JPEG	G -> B	R -> B	NCC G / R	L2 G / R
cathedral	(5, 2)	(12, 3)	0.95237 / 0.90071	0.30866 / 0.44563
monastery	(-3, 2)	(3, 2)	0.8405 / 0.66113	0.56481 / 0.82325
tobolsk	(3, 2)	(6, 3)	0.92034 / 0.85351	0.39915 / 0.54128

Why the crop is safer

The valid intersection is computed from the measured shifts, then border rows and columns are tested against central-scene statistics. A short consecutive-run rule avoids stopping at a single deceptively clean strip; a small final safety guard removes the colored scan frame that can sit immediately before a white run. No image-specific crop coordinates are encoded.

All 14 course-provided samples

The table documents the complete supplied dataset, not only the cover images. Positive dy moves a plate downward and positive dx moves it right. NCC is higher-is-better; normalized L2/RMSE is lower-is-better.

image	G -> B	R -> B	NCC G / R	L2 G / R	mean NCC / L2	crop t,b,l,r
cathedral	(5, 2)	(12, 3)	0.95237 / 0.90071	0.30866 / 0.44563	0.92654 / 0.37714	(17, 331, 23, 368)
church	(26, -2)	(55, -9)	0.81778 / 0.77351	0.60369 / 0.67304	0.79564 / 0.63837	(91, 3171, 136, 3447)
emir	(50, 22)	(106, 40)	0.65608 / 0.32704	0.82936 / 1.16014	0.49156 / 0.99475	(169, 3076, 303, 3492)
harvesters	(61, 17)	(124, 14)	0.72342 / 0.78307	0.74375 / 0.65869	0.75324 / 0.70122	(221, 3155, 223, 3413)
icon	(41, 16)	(90, 23)	0.71687 / 0.67447	0.75251 / 0.80688	0.69567 / 0.77969	(121, 3165, 280, 3529)
ilemselga	(43, 2)	(129, 9)	0.75468 / 0.64177	0.70046 / 0.84644	0.69822 / 0.77345	(160, 3224, 393, 3510)
melons	(83, 6)	(179, 13)	0.85247 / 0.91101	0.54319 / 0.42188	0.88174 / 0.48253	(293, 3212, 235, 3459)
monastery	(-3, 2)	(3, 2)	0.8405 / 0.66113	0.56481 / 0.82325	0.75081 / 0.69403	(18, 311, 29, 369)
religious_painting	(27, 3)	(68, 8)	0.69725 / 0.65343	0.77814 / 0.83255	0.67534 / 0.80534	(423, 3020, 212, 3449)
self_portrait	(80, 29)	(176, 36)	0.64813 / 0.70996	0.83889 / 0.76163	0.67905 / 0.80026	(219, 3162, 411, 3482)
siren	(51, -7)	(96, -25)	0.70208 / 0.86407	0.7719 / 0.52141	0.78308 / 0.64665	(224, 3154, 288, 3537)
three_generations	(50, 6)	(112, 10)	0.83523 / 0.87911	0.57406 / 0.49171	0.85717 / 0.53288	(261, 3163, 228, 3504)
tobolsk	(3, 2)	(6, 3)	0.92034 / 0.85351	0.39915 / 0.54128	0.88692 / 0.47022	(13, 326, 30, 372)
wharf	(15, -8)	(83, -17)	0.8637 / 0.81148	0.52211 / 0.61403	0.83759 / 0.56807	(176, 3205, 191, 3429)

The fixed method is deliberately global: low scores on a scene with large sky, water, or historical exposure differences are reported rather than hidden. The visual outputs remain the primary reconstruction result; the metrics make the search auditable.

Additional Library of Congress examples

Deliverable requirement: results on 3+ self-selected examples downloaded from the Prokudin-Gorskii collection. These are three original LOC 'digital file from glass neg.' vertical TIFF resources. They were downloaded, split into B / G / R thirds, and processed by exactly the same code and parameters as the 14 course samples.



V lasnoe polianie



V lasnoe polianie (variant)



Na Uralie

item	title	NCC G / R	L2 G / R	mean NCC / L2	offsets G / R
LOC 2018679023	V lasnoe polianie	0.77135 / 0.81494	0.67624 / 0.60837	0.79314 / 0.64231	[42, -6] / [102, 7]
LOC 2018679024	V lasnoe polianie (variant)	0.80382 / 0.77942	0.62638 / 0.6642	0.79162 / 0.64529	[38, 6] / [73, 14]
LOC 2018679025	Na Uralie	0.70154 / 0.83208	0.77261 / 0.57952	0.76681 / 0.67606	[41, 34] / [95, 58]

LOC item 2018679023: <https://www.loc.gov/item/2018679023/>

Raw plate: <https://tile.loc.gov/storage-services/master/pnp/prok/00100/00159u.tif>

LOC item 2018679024: <https://www.loc.gov/item/2018679024/>

Raw plate: <https://tile.loc.gov/storage-services/master/pnp/prok/00100/00160u.tif>

LOC item 2018679025: <https://www.loc.gov/item/2018679025/>

Raw plate: <https://tile.loc.gov/storage-services/master/pnp/prok/00100/00161u.tif>

Visual appendix - all reconstructed outputs

The HTML page provides clickable full-size images. This appendix confirms that all 14 course outputs were generated by the same run and keeps the report self-contained for PDF review.



cathedral



tobolsk



melons



three_generations



wharf



church



siren



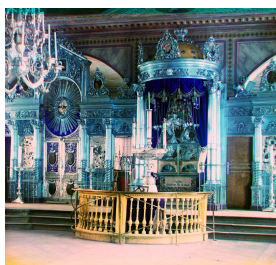
harvesters



monastery



ilemselga



icon



self_portrait



religious_painting



emir

Source and reproducibility

Course inputs are the supplied CS180_fa2026_proj1_data files. The three extra plates are linked above to LOC item pages and exact TIFF resources. The implementation is code/main.py; the browser report is web/index.html; the Gradescope artifact is this PDF.